

Project Design Phase-II
Solution Requirements (Functional & Non-Functional)

Date	10 July 2026
Team ID	
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	4 Marks

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement
FR-1	Loan Application Submission	Applicant fills the Predict page form with personal and financial details.
FR-2	Data Preprocessing	Categorical fields label-encoded; numerical fields scaled with StandardScaler for KNN.
FR-3	Prediction & Decision	KNN model predicts Loan_Status (Approved / Rejected) from the encoded, scaled input.
FR-4	Result Explanation	Application logic derives eligibility %, confidence score, risk level, reasons and recommendations from the prediction probability.
FR-5	Applicant Profile	Display the applicant's submitted information on a dedicated profile page.
FR-6	History Management	Store and display previous prediction records.
FR-7	Informational Pages	Home page introduces the platform; About page documents the project.

Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple, responsive form-driven UI built with HTML5, CSS3, JavaScript and Font Awesome.
NFR-2	Performance	Prediction response returned within a few seconds of form submission.
NFR-3	Reliability	Model, encoder and scaler are versioned .pkl artifacts loaded consistently at runtime.
NFR-4	Maintainability	Flask backend structured to allow swapping the deployed model after re-training.
NFR-5	Portability	Runs on any environment with Python, Flask and the required ML libraries installed.